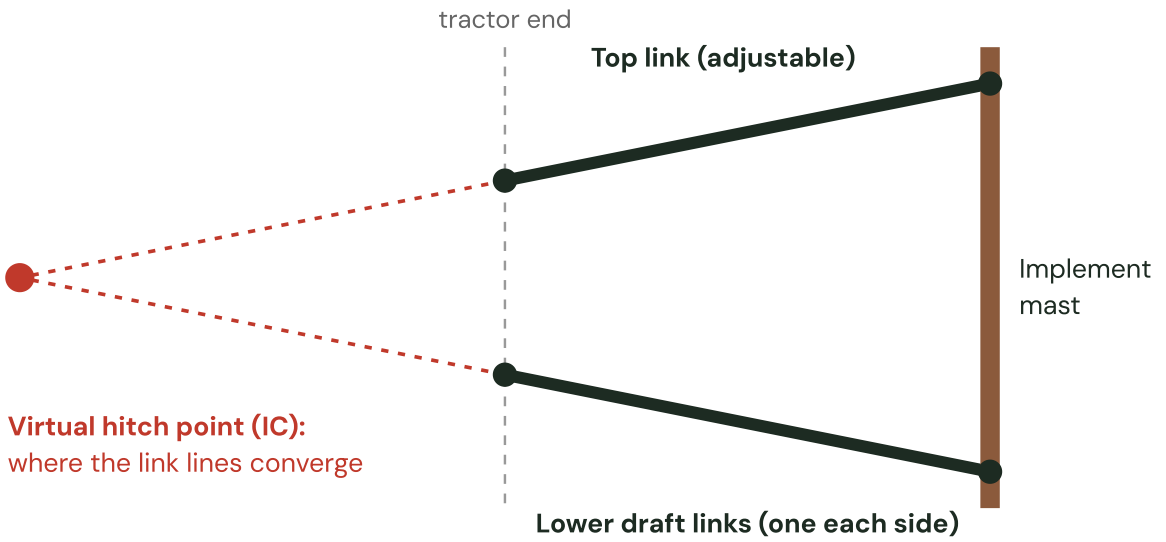
UNIT 4 · TOPIC 9(B)

Hitching Systems and the Three-Point Linkage

Syllabus topic: Tractor chassis mechanics and hitching systems



Mounted: full weight transfer • Semi-mounted: partial • Trailed: drawbar only

Three-point hitching system, side view (redrawn from the source figure)

Three-point linkage

- TWO lower draft links (lifted by the hydraulic lift arms via lift rods) + ONE adjustable top link.
- The implement mast completes a rigid triangle when raised.

Category (IS 4468 / ISO 730)	Lower hitch pin	Top pin	Mast height
Cat-I — small/medium tractors	~22 mm	~19 mm	460 mm
Cat-II	~28 mm	~25.5 mm	610 mm
Cat-III / IV	Large tractors	—	—

ISO 730 assigns Category 1 to tractors of up to 48 kW (the source figure gave "≤ ~48 hp", a unit slip). Check ball/pin sizes when matching implements (Unit 3, Topic 13).

Hitching types

Type	How carried	Weight transfer
MOUNTED	Fully carried on the linkage	Best; compact turning
SEMI-MOUNTED	Front on the links, rear on its own wheel	Partial
TRAILED	Drawbar only	Least

Geometry and adjustments

- The lower links converge in plan and elevation → the **VIRTUAL HITCH POINT**, which governs force analysis and stability (Unit 2, Topic 5).
- Sway chains/stabilisers limit lateral swing.
- The levelling lever (right lift rod) sets lateral level.
- Top-link length sets pitch (share-point "suction" attitude).

Practice MCQs

Q149. The three-point linkage consists of:

(a) 3 top links
(b) Two lower draft links + one top link
(c) One chain
(d) 4 arms

Q150. Category-I linkage mast height is:

(a) 360 mm
(b) 460 mm
(c) 610 mm
(d) 810 mm

Q151. Category-II lower hitch pins are about:

(a) 16 mm
(b) 22 mm
(c) 28 mm
(d) 36 mm

Q152. Hitch categories are specified in:

(a) IS 6284
(b) IS 4468 / ISO 730
(c) IS 9020
(d) ISO 500

Q153. The virtual hitch point of a mounted implement is the:

(a) Drawbar pin
(b) Convergence point of the lower links
(c) Top-link ball
(d) PTO end

Q154. Best weight transfer to the tractor is given by which implement attachment?

(a) Trailed
(b) Semi-mounted
(c) Fully mounted
(d) Winched

Q155. Lateral levelling of a mounted implement is set by the:

(a) Top link
(b) Right lift-rod levelling lever
(c) Sway chain
(d) Drawbar

Q156. Implement pitch (share suction attitude) is set by the:

(a) Top-link length
(b) Tyre pressure
(c) Seat
(d) PTO

Answers

Q149 — (b) Two lower draft links + one top link
Q150 — (b) 460 mm
Q151 — (c) 28 mm
Q152 — (b) IS 4468 / ISO 730
Q153 — (b) Convergence point of the lower links
Q154 — (c) Fully mounted
Q155 — (b) Right lift-rod levelling lever
Q156 — (a) Top-link length

Sources: Liljedahl et al. — Tractors and their Power Units; Kepner et al. (hitching); IS 4468/ISO 730; Jagdishwar Sahay; NPTEL lectures.
